Hydraulic Valve Operation in CO_2 Cleaning Processes and Gasket Material Selection for Pressure Resistance, Thermal Stability, and Service Life

Abstract

Hydraulic valves are critical components in CO_2-based cleaning systems, where they must operate reliably under elevated pressures (7–30 MPa) and temperatures (31–150°C) characteristic of supercritical CO_2 processing. Gasket failure represents a primary mode of valve malfunction in these systems, driven by CO_2 sorption, plasticization, and rapid gas decompression (RGD) damage in elastomeric materials. This paper presents a systematic framework for hydraulic valve operation in CO_2 cleaning processes and gasket material selection based on pressure resistance, thermal stability, and service life criteria. Based on comparative studies of elastomer performance in high-pressure CO_2 environments, hydrogenated nitrile butadiene rubber (HNBR) exhibits the highest resistance to rapid decompression damage, with maximum logarithmic strain of only 0.17 under 4 MPa CO_2 pressure, compared to 1.42 for EPDM. HNBR demonstrates superior performance for CO_2 valve applications, while PTFE-based materials offer excellent chemical inertness for static sealing. A service life framework is proposed based on operating pressure, temperature, decompression rate, and material selection, with maintenance intervals of 6–12 months recommended for elastomeric gaskets in CO_2 service.

Keywords: Hydraulic valve; CO_2 cleaning; Gasket material; Rapid gas decompression; HNBR; PTFE; Service life

1. Introduction

Supercritical CO_2 ($scCO_2$) cleaning has emerged as an environmentally friendly, highly efficient technique for precision cleaning in the food, pharmaceutical, cosmetic, and electronics industries . The process exploits the unique properties of CO_2 above its critical point (31.1°C, 7.38 MPa), where it exhibits gas-like diffusivity and liquid-like density, enabling deep penetration into complex geometries and effective removal of organic residues .

Hydraulic valves in CO_2 cleaning systems regulate the flow of supercritical or liquid CO_2 between the extraction vessel, separator, and circulation loop. These valves must operate reliably under conditions that challenge both metallic and non-metallic components: pressures ranging from 7 to 30 MPa, temperatures from 31 to 150°C, and frequent pressure cycling during batch processing .

Valve gaskets are the weakest link in these systems. Unlike metallic valve components, elastomeric and polymeric gaskets are susceptible to CO_2 sorption, which can lead to plasticization, swelling, and—under rapid decompression conditions—catastrophic failure through explosive decompression . Understanding the failure mechanisms and selecting appropriate gasket materials is therefore essential for reliable valve operation.

This paper addresses two interconnected objectives: (1) establishing the operational requirements for hydraulic valves in CO_2 cleaning processes, and (2) developing a material selection framework for valve gaskets based on pressure resistance, thermal stability, and service life criteria.

2. Operational Requirements for Hydraulic Valves in CO_2 Cleaning Systems

2.1 System Configuration and Valve Function

A typical supercritical CO_2 cleaning system comprises a CO_2 supply, a pressure vessel (cleaning chamber), a circulation pump, a separator, and associated piping. Hydraulic valves serve several critical functions:

- **Isolation valves**: Isolate the cleaning vessel during loading/unloading
- **Pressure control valves**: Regulate system pressure during cleaning cycles
- **Flow control valves**: Modulate CO_2 flow rate through the cleaning circuit
- **Depressurization valves**: Controlled release of CO_2 after cleaning

The cleaning process itself involves sequential stages: pressurization to supercritical conditions, static cleaning (1–2 hours), dynamic cleaning (2–4 hours), and controlled depressurization .

2.2 Pressure and Temperature Conditions

Operational parameters for CO_2 cleaning systems are well-established in the literature. The cleaning vessel is typically maintained at **31–150°C** and **7–30 MPa** to achieve supercritical conditions . For typical operation, **40–60°C and 15–25 MPa** are recommended . These conditions define the minimum requirements for valve gasket materials.

Table 1: Typical Operating Parameters for CO_2 Cleaning System Valves

Parameter	Range	Typical Value
Pressure	7–30 MPa	15–25 MPa
Temperature	31–150°C	40–60°C
Flow rate	5–10 L/min	5–10 L/min
Cleaning cycle	2–6 hours	2–4 hours
Depressurization rate	Controlled	Must avoid rapid decompression

2.3 Valve Operation During Cleaning Cycles

Proper valve operation is critical for both cleaning efficiency and equipment longevity. Key operational considerations include:

Pressurization: CO_2 should be introduced gradually to avoid pressure surges that could damage gaskets. The system should be pressurized to the target pressure while monitoring for leaks.

Cleaning phase: Valves should maintain stable pressure and flow throughout the static and dynamic cleaning phases. Any valve leakage during this phase reduces cleaning efficiency and contaminates the CO_2 stream.

Depressurization: This is the most critical phase for gasket life. **Rapid decompression must be avoided** . The depressurization rate directly influences the extent of RGD damage in elastomeric gaskets: slow depressurization rates show significantly less damage compared to fast rates .

2.4 Cleaning and Maintenance of Valve Components

CO_2 cleaning systems require regular maintenance to ensure reliable operation. Seals should be inspected for wear or aging and replaced as necessary . For new equipment or units that have been idle, three cleaning cycles with pure ethanol followed by supercritical CO_2 purging are recommended .

Maintenance intervals for valve components in CO_2 service are typically:

- **Visual inspection**: Every 4 weeks
- **Seal/gasket replacement**: Every 6–12 months depending on usage intensity
- **Full valve overhaul**: Every 12–24 months

A Polish service provider reports receiving dozens of monthly service requests where the cause is lack of regular cleaning or worn gaskets, recommending gasket replacement **at least once every 12 months**, and **every 6 months** for intensively used equipment .

3. Gasket Material Selection Framework

3.1 Failure Mechanisms in CO_2 Service

Gasket materials in CO_2 cleaning systems face three primary failure mechanisms:

Plasticization and Swelling: CO_2 absorption into polymer matrices causes plasticization, lowering glass transition temperature (Tg) and reducing mechanical strength. CO_2 at weight concentrations of 8–10% can plasticize common glassy polymers with Tg in the 8–100°C range to below room temperature . Polymers with flexible backbones, high free volume, and low Tg (characteristic of elastomers) exhibit higher CO_2 solubility .

Rapid Gas Decompression (RGD): When pressure is released, dissolved CO_2 becomes supersaturated and nucleates as gas bubbles within the polymer. This causes internal blistering, cracking, and potentially catastrophic fragmentation . RGD is the most destructive failure mode for elastomeric seals in CO_2 service. The S/D ratio (solubility coefficient/diffusion coefficient) determines RGD susceptibility, and this ratio is **highest for CO_2 among common gases**, meaning polymers exhibit maximum CO_2 effects .

Chemical Degradation: CO_2 is generally chemically inert toward most polymers, but moisture and contaminants can induce chemical changes. FTIR analysis has identified significant chemical shifts in some elastomers (notably Neoprene) after CO_2 exposure .

3.2 Comparative Performance of Gasket Materials

A comprehensive study evaluated four sealing materials—HNBR, NBR, EPDM, and NR—under high-pressure CO_2 conditions using a coupled diffusion-deformation model . The results provide quantitative selection criteria:

Table 2: Comparative Performance of Elastomeric Gasket Materials in High-Pressure CO_2

Material	Max Logarithmic Strain (4 MPa)	RGD Resistance Ranking	Notes
HNBR	0.17	1 (Best)	Superior RGD resistance
NR	—	2	Good performance
NBR	—	3	Moderate performance
EPDM	1.42	4 (Worst)	Exceeds elongation at break

Under 4 MPa external pressure, **HNBR exhibits the best resistance to rapid decompression**, with maximum logarithmic strain of only 0.17, while EPDM reaches a strain peak of 1.42—exceeding its elongation at break and resulting in rupture . The material resistance ranking is **HNBR > NR > NBR > EPDM** .

Compression ratio also influences performance. For NBR, EPDM, and NR, moderately increasing compression ratio reduces strain during rapid decompression . For HNBR, the strain influence pattern occurs earlier than for other materials due to its unique cavity pressure differential characteristics .

Thermoplastic polymers show less CO_2 effect than elastomers overall . Among thermoplastics, **PTFE (Teflon)** is widely used in valve seats and pressure boundary connections in sCO_2 systems . PTFE's C-F bonds appear capable of slowing CO_2 diffusion through free volume while enabling weak Van der Waals interactions between fluorine and CO_2 . However, PTFE does exhibit some striations perpendicular to the O-ring axis after prolonged sCO_2 exposure .

3.3 Material Selection Recommendations

Based on the comparative data, the following selection framework is proposed:

Primary Recommendation: HNBR

HNBR (hydrogenated nitrile butadiene rubber) is the optimal choice for dynamic sealing applications in CO_2 cleaning valves. Its superior RGD resistance (strain 0.17 at 4 MPa) and favorable ranking among elastomers make it the default selection for valve stem seals and dynamic gaskets .

Alternative: PTFE for Static Sealing

For static gasket applications where chemical inertness and thermal stability are paramount, PTFE-based materials offer excellent performance. PTFE's C-F bonds provide chemical resistance and slow CO_2 diffusion . PTFE is commonly used in valve seats and pressure boundary connections .

Caution: EPDM

EPDM should be avoided for high-pressure CO_2 service. Its high strain under rapid decompression (1.42) exceeds its elongation at break, leading to rupture . EPDM's swelling behavior (400% for NR vs. 70% for EPDM as filler content increases) suggests it is more susceptible to CO_2-induced dimensional changes .

3.4 Effect of Reinforcement and Fillers

The role of reinforcing fillers has been systematically analyzed. Reinforced HNBR and FKM materials were found to suffer damage after return to zero pressure at both 60°C and 130°C (6 MPa exposure), inducing loss of rigidity and making them **unsuitable for gaseous environments** . This suggests that **unfilled or lightly filled elastomers** may perform better in CO_2 service than heavily reinforced grades.

For PTFE, filler content affects compression performance. Filler concentrations of 20–30 vol% represent the optimal window for balancing compression strength and processability.

4. Pressure Resistance and Thermal Stability

4.1 Pressure Resistance Requirements

Valve gaskets in CO_2 cleaning systems must withstand:

- **Static pressure**: 7–30 MPa continuous
- **Cyclic pressure**: Repeated pressurization/depressurization cycles
- **Pressure differentials**: Across the gasket during valve operation

The primary concern is not the absolute pressure (which is within the capability of most engineering polymers) but rather the **decompression behavior**. RGD damage occurs during pressure release, not during sustained pressure exposure.

Key finding: Compression ratio significantly affects RGD resistance. A **moderate compression proportion of approximately 15%** is beneficial for prolonging O-ring fatigue life while minimizing leakage risks .

4.2 Thermal Stability

Operating temperatures of 40–60°C are typical, with excursions up to 150°C possible during certain cleaning protocols . Thermal stability requirements are therefore moderate for most elastomers, but the **combination of elevated temperature and CO_2 exposure** accelerates degradation.

HNBR and FKM were tested at 60°C and 130°C under 6 MPa CO_2. HNBR was identified as the most appropriate material in this context . At 100°C, both FF-202 (a perfluoroelastomer) and EPDM performed well; at 150°C, FF-202 showed better performance .

For PTFE, thermal stability is excellent (continuous use up to 260°C), and the material is widely used in sCO_2 systems at elevated temperatures .

5. Service Life Framework

5.1 Factors Affecting Gasket Service Life

The service life of valve gaskets in CO_2 cleaning systems depends on:

1. **Material selection**: HNBR >> NBR > EPDM for RGD resistance
2. **Operating pressure**: Higher pressure increases CO_2 sorption and RGD risk
3. **Decompression rate**: Faster decompression dramatically increases damage
4. **Temperature**: Higher temperature increases CO_2 diffusivity but may reduce solubility
5. **Compression ratio**: Moderate compression (~15%) optimizes fatigue life
6. **Cavity size**: Larger cavities reduce fatigue life; cavity-containing O-rings have approximately **1/25 the fatigue life** of cavity-free O-rings

5.2 Quantitative Service Life Estimates

Based on fatigue modeling for rubber O-rings in high-pressure CO_2:

- As CO_2 pressure increases from **1 MPa to 5 MPa**, fatigue life of O-rings with cavities decreases from **infinity to 7.36 cycles**

- Acceleration of depressurization rate adversely affects fatigue life

- Cavity diameter increase reduces fatigue life

For practical service life estimation in CO_2 cleaning systems (15–25 MPa operating pressure), these data suggest that gaskets should be considered **consumable items** with replacement intervals of **6–12 months** depending on usage intensity .

5.3 Maintenance and Replacement Schedule

A recommended maintenance framework for CO_2 cleaning system valve gaskets:

Table 3: Recommended Gasket Maintenance Schedule

Activity	Interval	Notes
Visual inspection	4 weeks	Check for leaks, deformation, cracks
Functional test	Monthly	Manual valve operation
Gasket replacement (intensive use)	6 months	High cycle frequency
Gasket replacement (normal use)	12 months	Standard operation
Full valve overhaul	12–24 months	Complete disassembly and inspection

This schedule is consistent with industry practice for CO_2 equipment maintenance .

6. Conclusions

Hydraulic valve operation in CO_2 cleaning systems requires careful attention to gasket material selection and depressurization procedures. The following conclusions are drawn:

1. **HNBR is the optimal gasket material** for dynamic sealing in CO_2 cleaning valves, exhibiting superior rapid decompression resistance (strain 0.17 at 4 MPa) compared to EPDM (strain 1.42, resulting in rupture).

2. **PTFE-based materials** are recommended for static sealing applications where chemical inertness and thermal stability are critical, though some surface texturization occurs after prolonged sCO_2 exposure.

3. **EPDM should be avoided** in high-pressure CO_2 service due to its poor RGD resistance and high swelling tendency.

4. **Reinforced elastomers may be unsuitable** for gaseous CO_2 environments, as they can suffer damage and loss of rigidity after pressure cycling.

5. **Rapid decompression must be avoided**; controlled depressurization is essential for maximizing gasket service life.

6. **A moderate compression ratio of approximately 15%** optimizes fatigue life while maintaining sealing effectiveness.

7. **Gasket replacement intervals of 6–12 months** are recommended, with 6-month intervals for intensive use and 12-month intervals for normal operation.

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